

9. Ascorbic acid, $C_6H_8O_6$, is the main component of vitamin C tablets. Its name is derived from *a-* (meaning “no”) and *scorbutus* (scurvy), the disease caused by a deficiency of vitamin C. A student was asked to find the percentage of ascorbic acid in identical vitamin C tablets.

She was told to use the following method.

- Fill a burette with $0.100 \text{ mol dm}^{-3}$ sodium hydroxide solution.
- Weigh a conical flask and record its mass.
- Add a vitamin C tablet to the flask, reweigh it and record its mass.
- Add about 50 cm^3 of deionised water to the flask and swirl to break up the tablet.
- Heat the flask gently for 5 to 10 minutes.
- After the solution has cooled add a few drops of a suitable indicator.
- Carry out a rough titration of this solution with the sodium hydroxide solution.
- Accurately repeat the procedure several times and calculate a mean titre.

- (a) A **three** decimal place balance was used. The mass of each vitamin C tablet was 500 mg.

Calculate the maximum percentage error in the weighing of the tablet.
You **must** show your working.

[2]

Maximum percentage error = %

- (b) (i) Suggest why she did not need to measure the volume of water accurately. [1]

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- (ii) Suggest why she heated the flask for 5 to 10 minutes. [1]

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- (c) The student used the results from three titrations to calculate a mean titre. Some of her results are shown below.

Titration	1	2	3
Final reading / cm ³	26.90	26.90	
Initial reading / cm ³	0.25	0.15	0.20
Titre / cm ³	26.65	26.75	

Mean titre = 26.73 cm³

Determine the **final reading** for the third titration.

[2]

Final reading = cm³

- (d) Ascorbic acid can decompose upon exposure to air. If this reaction occurred before the titration was completed, state how it might affect the titration results. Explain your answer. [2]

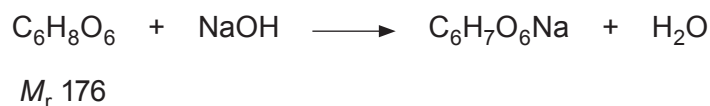
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- (e) The equation for the reaction between ascorbic acid and sodium hydroxide is given below.



The percentage of ascorbic acid is identical in each 500 mg tablet. Calculate the percentage of ascorbic acid in each vitamin C tablet. [3]

Percentage ascorbic acid = %

- (f) Sulfuric acid and hydrochloric acid are strong acids.

- (i) Calculate the pH of a solution of $0.010 \text{ mol dm}^{-3}$ sulfuric acid, H_2SO_4 . [2]

pH =

- (ii) When hydrochloric acid is heated with MnO_2 it reacts according to the following equation.



Explain why this can be classified as a redox reaction. [2]

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